

Microeconomics

Lecture 3: Budget Constraint and Optimal Choice

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What We Know So Far

- **Consumer preferences** → rationality axioms
- **Utility function** $U(x, y)$ → numerical representation
- **Indifference curves** → graphical representation
- **MRS** = MU_x/MU_y → subjective trade-off between goods

Open question:

*The consumer knows what she wants, but **what can she afford?***

Budget Constraint

Today we introduce the **budget constraint** (what the consumer can buy) and learn how to find the **optimal choice** that maximizes utility given the constraint.

Lecture Objectives

After this lecture you will be able to:

1. **Write and interpret** the budget constraint
2. **Graph** the budget line
3. **Analyze** the effect of changes in prices and income
4. **Set up** the consumer's optimal choice problem
5. **Solve** the problem with the substitution and Lagrangian methods
6. **Compute** the optimal choice for Cobb-Douglas functions

Lecture Focus

By the end of the lecture you will be able to find exactly which bundle (x^*, y^*) maximizes the consumer's utility, given prices and income.

The Fundamental Economic Problem

Realistic situation:

- The consumer has preferences (Lecture 2: indifference curves)
- She would like to consume as much as possible of everything
- **But:** she has limited resources (finite income)
- She must pay market prices for the goods

Hence:

1. Which bundles (x, y) can she afford? $\rightarrow$ *budget constraint*
2. Which one will she choose among the affordable ones? $\rightarrow$ *optimal choice (utility maximization)*

Budget Constraint: Definition

Given:

- p_x = unit price of good x (€/unit)
- p_y = unit price of good y (€/unit)
- M = consumer's disposable income (€)
- (x, y) = quantities purchased of the two goods

Budget Constraint

Total expenditure cannot exceed income:

$$p_x \cdot x + p_y \cdot y \leq M$$

Interpretation:

- $p_x \cdot x$ = expenditure on good x
- $p_y \cdot y$ = expenditure on good y
- $p_x \cdot x + p_y \cdot y \leq M \rightarrow$ I cannot spend more than I have

Budget Line

Budget Line

Set of bundles that exactly exhaust income:

$$p_x x + p_y y = M$$

Explicit form (solving for y):

$$y = \frac{M}{p_y} - \frac{p_x}{p_y} x$$

Components:

- **Vertical intercept:** $y = M/p_y$ (when $x = 0$)
→ Maximum quantity of y affordable if I spend everything on y
- **Horizontal intercept:** $x = M/p_x$ (when $y = 0$)
→ Maximum quantity of x affordable if I spend everything on x
- **Slope:** $-p_x/p_y$ (coefficient of x)
→ Rate at which I can substitute x for y in the market

Example: Pizzas and Beers

- $M = 60$ euros (disposable income)
- $p_x = 10$ euros (price of a pizza)
- $p_y = 3$ euros (price of a beer)

Budget constraint:

$$10x + 3y = 60$$

Explicit form:

$$y = \frac{60}{3} - \frac{10}{3}x = 20 - \frac{10}{3}x$$

Intercepts:

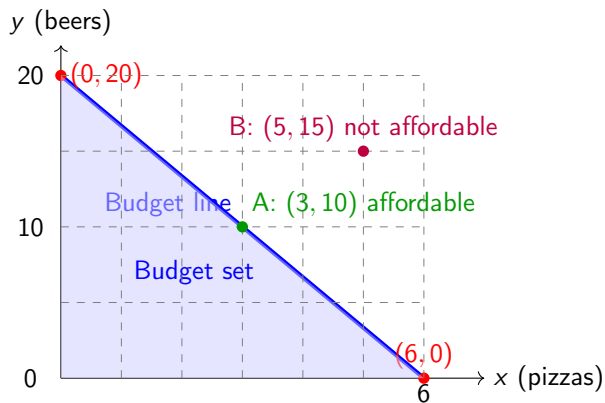
- Maximum pizzas (if $y = 0$): $x = 60/10 = 6$ pizzas
- Maximum beers (if $x = 0$): $y = 60/3 = 20$ beers

Slope: $-10/3 \approx -3.33$

→ To buy 1 more pizza, I must give up 3.33 beers

Graph: Budget Line

Example: $10x + 3y = 60$



Note: The y axis is **compressed** for display (real economic coordinates: y up to 20 beers).

Budget Set

Definition

The **budget set** is the set of all affordable (purchasable) bundles:

$$B = \{(x, y) : p_x x + p_y y \leq M, x \geq 0, y \geq 0\}$$

Components:

- **Frontier:** Budget line ($p_x x + p_y y = M$)
→ Bundles that exhaust all income
- **Interior:** Bundles below the line ($p_x x + p_y y < M$)
→ Bundles with expenditure below income (positive saving)
- **Non-negativity constraints:** $x \geq 0, y \geq 0$
→ I cannot buy negative quantities

Geometric shape: Triangle in the first quadrant with vertices:

- Origin: $(0, 0)$
- $(M/p_x, 0)$ on the x axis
- $(0, M/p_y)$ on the y axis

Interpreting the Slope

Slope of the budget line: $-p_x/p_y$

Market Rate of Substitution

The slope represents the **objective trade-off** imposed by the market: how many units of y I must give up to buy 1 unit of x .

Example: $-p_x/p_y = -10/3$

- To buy 1 more pizza ($\Delta x = +1$):
- I must spend €10 more
- Reducing spending on beer: $10/3 \approx 3.33$ fewer beers
- Hence: $\Delta y = -10/3$ when $\Delta x = +1$

Comparison with the MRS:

- **MRS** = *subjective* trade-off (consumer preferences)
- p_x/p_y = *objective* trade-off (market prices)
- **Optimum:** when the two coincide! (see later)

Example: Three Bundles

Data: $M = 60$, $p_x = 10$, $p_y = 3$

Bundle A: (3, 10)

- Expenditure: $10 \cdot 3 + 3 \cdot 10 = 30 + 30 = 60 \checkmark$
- On the budget line $\rightarrow$ exhausts income

Bundle B: (2, 8)

- Expenditure: $10 \cdot 2 + 3 \cdot 8 = 20 + 24 = 44 < 60 \checkmark$
- Inside the budget set $\rightarrow$ saving €16

Bundle C: (5, 15)

- Expenditure: $10 \cdot 5 + 3 \cdot 15 = 50 + 45 = 95 > 60 \times$
- Outside the budget set $\rightarrow$ not affordable

Question: Which one will the consumer choose? $\rightarrow$ It depends on preferences!

Properties of the Budget Constraint

1. Linearity

- The budget line is always a straight line (no curves)
- Constant slope $-p_x/p_y$ everywhere
- Implication: constant market rate of substitution

2. Homogeneity of degree zero

- If I multiply (p_x, p_y, M) by the same constant $\lambda > 0$:
 $\lambda p_x x + \lambda p_y y = \lambda M \rightarrow$ budget set unchanged
- Example: I double all prices and income $\rightarrow$ no real change
- Implication: only *relative* prices and income matter

3. Convexity

- The budget set is convex (triangular shape)
- A weighted average of two affordable bundles is affordable
- Implication: well-behaved optimization problem

Summary of Part 1

1. Budget Constraint

- Equation: $p_x x + p_y y = M$
- Limits the bundles affordable to the consumer

2. Budget Line

- Explicit form: $y = M/p_y - (p_x/p_y)x$
- Intercepts: $(M/p_x, 0)$ and $(0, M/p_y)$
- Slope: $-p_x/p_y$ (market trade-off)

3. Budget Set

- All bundles with $p_x x + p_y y \leq M$
- Triangular shape in the first quadrant

Changes in the Constraint

Situation: We understand the budget constraint for given (p_x, p_y, M) .

Questions:

1. What happens if **income** M rises or falls?
2. What happens if a **price** p_x or p_y rises or falls?
3. What happens with **proportional inflation** (all prices rise)?

Goal:

- Understand how the budget line changes graphically
- Distinguish between **parallel shifts** and **rotations**
- Interpret each change economically

Increase in Income

Scenario: M rises to $M' > M$, prices unchanged.

New budget line:

$$y = \frac{M'}{p_y} - \frac{p_x}{p_y}x$$

Comparison with the original line:

- **Slope:** $-p_x/p_y$ (unchanged)
- **Vertical intercept:** $M'/p_y > M/p_y$ (rises)
- **Horizontal intercept:** $M'/p_x > M/p_x$ (rises)

Graphical effect: *Parallel outward shift*

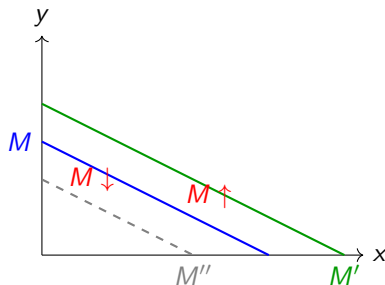
Economic interpretation:

- I can buy more of both goods
- The budget set expands
- The market trade-off (p_x/p_y) stays the same

Decrease in Income

Scenario: M falls to $M'' < M$, prices unchanged.

Effect: Parallel inward shift



General rule:

- $M \uparrow \rightarrow$ the line shifts outward (parallel)
- $M \downarrow \rightarrow$ the line shifts inward (parallel)
- Slope always unchanged

Increase in the Price of x

Scenario: p_x rises to $p'_x > p_x$, M and p_y unchanged.

New line:

$$y = \frac{M}{p_y} - \frac{p'_x}{p_y}x$$

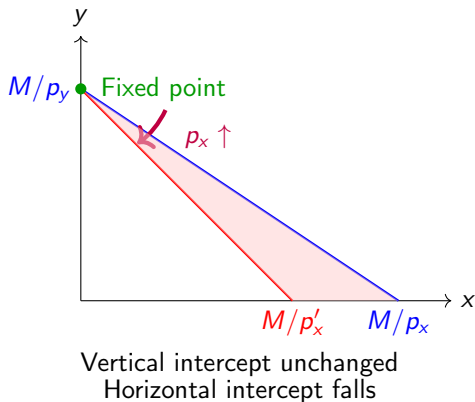
Comparison:

- **Vertical intercept:** M/p_y (unchanged)
→ If I buy only y , p_x does not matter
- **Horizontal intercept:** $M/p'_x < M/p_x$ (falls)
→ I can buy less x
- **Slope:** $-p'_x/p_y$ steeper (in absolute value)
→ x has become relatively more expensive

Graphical effect: Inward rotation around the vertical intercept

Graph: Change in p_x

Increase in p_x : Inward rotation



Interpretation: x more expensive $\rightarrow$ I can buy less x (other things equal)

Decrease in the Price of x

Scenario: p_x falls to $p_x'' < p_x$, M and p_y unchanged.

Effect: Outward rotation around the vertical intercept

- Vertical intercept: unchanged
- Horizontal intercept: $M/p_x'' > M/p_x$ (rises)
- Slope: flatter (in absolute value)

Similarly for changes in p_y :

- $p_y \uparrow$: Inward rotation around the horizontal intercept
- $p_y \downarrow$: Outward rotation around the horizontal intercept

General rule:

- Price rises $\rightarrow$ inward rotation
- Price falls $\rightarrow$ outward rotation
- Fixed point: intercept of the good whose price has NOT changed

Proportional Inflation

Scenario: All prices and income rise by the same factor $\lambda > 1$:

$$p'_x = \lambda p_x, \quad p'_y = \lambda p_y, \quad M' = \lambda M$$

New budget line:

$$\lambda p_x x + \lambda p_y y = \lambda M$$

Dividing by λ :

$$p_x x + p_y y = M$$

→ **Identical to the original line!**

Homogeneity of Degree Zero

Proportional inflation **does not change** the real budget set. Only *relative* prices and income matter for the consumer's choices.

Implication: The consumer does not suffer from “money illusion”: she reacts to real, not nominal, changes.

Example: 50% Inflation

Original situation:

- $M = 60$ euros, $p_x = 10$ euros, $p_y = 3$ euros
- Line: $10x + 3y = 60 \rightarrow y = 20 - (10/3)x$

After 50% inflation ($\lambda = 1.5$):

- $M' = 90$ euros, $p'_x = 15$ euros, $p'_y = 4.5$ euros
- Line: $15x + 4.5y = 90$

Dividing by 1.5:

$$10x + 3y = 60$$

Identical to the original line! ✓

Checking the intercepts:

- Original: $(6, 0)$ and $(0, 20)$
- After inflation: $(90/15, 0) = (6, 0)$ and $(0, 90/4.5) = (0, 20)$ ✓

Summary Table: Changes

Change	Graphical Effect	Interpretation
$M \uparrow$	Parallel outward shift	More purchasing power
$M \downarrow$	Parallel inward shift	Less purchasing power
$p_x \uparrow$	Inward rotation (fixed on y axis)	x relatively more expensive
$p_x \downarrow$	Outward rotation (fixed on y axis)	x relatively cheaper
$p_y \uparrow$	Inward rotation (fixed on x axis)	y relatively more expensive
$p_y \downarrow$	Outward rotation (fixed on x axis)	y relatively cheaper
Inflation λ on everything	No change	Only relative prices matter

Exercise: Identify the Change

Original line: $5x + 2y = 100$

Question: Which change produces each new line?

a) $5x + 2y = 150$

Answer: M increased from 100 to 150 (parallel outward shift) ✓

b) $10x + 2y = 100$

Answer: p_x doubled (from 5 to 10), inward rotation around $(0, 50)$ ✓

c) $10x + 4y = 200$

Answer: 100% inflation: all prices and M doubled → **same line!** ✓

d) $5x + 4y = 100$

Answer: p_y doubled (from 2 to 4), inward rotation around $(20, 0)$ ✓

Summary of Part 2

1. Changes in Income

- $M \uparrow$ or $M \downarrow \rightarrow$ parallel shift (slope unchanged)

2. Changes in Prices

- p_x changes $\rightarrow$ rotation around the y intercept
- p_y changes $\rightarrow$ rotation around the x intercept
- Price $\uparrow \rightarrow$ inward rotation (less affordable)
- Price $\downarrow \rightarrow$ outward rotation (more affordable)

3. Proportional Inflation

- If (p_x, p_y, M) are multiplied by $\lambda \rightarrow$ set unchanged
- Only *relative* prices/income matter

The Consumer's Optimal Choice

- The consumer has preferences: $U(x, y)$ (indifference curves)
- She must satisfy the constraint: $p_x x + p_y y = M$ (budget line)

Question: Which bundle (x^*, y^*) will she choose?

Constrained Maximization Problem

The consumer chooses the bundle that:

- Maximizes her utility
- Among all affordable ones (in the budget set)

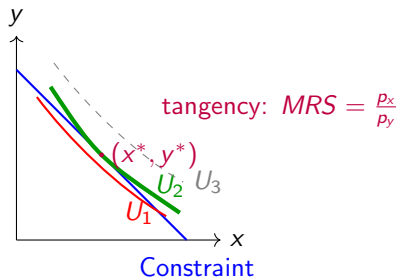
Formally:

$$\begin{aligned} \max_{x, y} \quad & U(x, y) \\ \text{s.t.} \quad & p_x x + p_y y = M \\ & x \geq 0, \quad y \geq 0 \end{aligned}$$

Graphical Solution: Tangency Condition

Intuitive idea:

- I want to reach the highest possible indifference curve
- But I must stay within the budget set
- Optimal solution: where the indifference curve is **tangent** to the budget line



U_1 : Affordable but not optimal (I can do better)

U_2 : Tangent $\rightarrow$ optimum!

U_3 : Too high, outside the budget

Mathematical Condition: Tangency

At the optimal point (x^*, y^*) :

Tangency Condition

The slope of the indifference curve must equal the slope of the budget line:

$$MRS = \frac{p_x}{p_y}$$

In terms of marginal utilities:

$$\frac{MU_x}{MU_y} = \frac{p_x}{p_y} \longrightarrow \frac{MU_x}{p_x} = \frac{MU_y}{p_y}$$

Equimarginal Principle

At the optimum, the **marginal utility per euro spent** must be equal for both goods.

Intuition: If $MU_x/p_x > MU_y/p_y$, it pays to shift €1 from y to x to raise total utility.

Why Tangency?

Proof by contradiction:

Suppose that at the optimum $MRS > p_x/p_y$ (e.g., $MRS = 3$, $p_x/p_y = 2$):

- $MRS = 3$: The consumer is willing to give up 3 units of y for 1 of x
- $p_x/p_y = 2$: The market requires only 2 units of y for 1 of x
- **Conclusion:** The consumer can do better by buying more x and less y !
→ The point was not optimal

Similarly, if $MRS < p_x/p_y$:

- The consumer can do better by buying more y and less x

Only when $MRS = p_x/p_y$:

- No marginal trade can increase utility
- → Local optimum reached

Method 1: Substitution

Problem:

$$\max_{x,y} U(x,y) \quad \text{s.t.} \quad p_x x + p_y y = M$$

Strategy:

1. From the constraint, express y as a function of x :

$$y = \frac{M - p_x x}{p_y}$$

2. Substitute into $U(x, y) \rightarrow$ function of x only:

$$\tilde{U}(x) = U\left(x, \frac{M - p_x x}{p_y}\right)$$

3. Maximize with respect to x :

$$\frac{d\tilde{U}}{dx} = 0$$

4. Solve for x^*
5. Substitute x^* into the constraint to find y^*

Example: Substitution with Cobb-Douglas

Problem: $U(x, y) = x^{0.5}y^{0.5}$, constraint $10x + 5y = 100$

Step 1: Constraint $\rightarrow y = (100 - 10x)/5 = 20 - 2x$

Step 2: Substitute into U :

$$\tilde{U}(x) = x^{0.5}(20 - 2x)^{0.5}$$

Step 3: Differentiate and set equal to zero:

$$\frac{d\tilde{U}}{dx} = 0.5x^{-0.5}(20 - 2x)^{0.5} + x^{0.5} \cdot 0.5(20 - 2x)^{-0.5} \cdot (-2) = 0$$

Simplifying (multiplying by $2x^{0.5}(20 - 2x)^{0.5}$):

$$(20 - 2x) - 2x = 0 \quad \Rightarrow \quad 20 = 4x \quad \Rightarrow \quad x^* = 5$$

Step 4: $y^* = 20 - 2(5) = 10$

Optimal solution: $(x^*, y^*) = (5, 10) \checkmark$

Method 2: Lagrangian

Problem:

$$\max_{x,y} U(x,y) \quad \text{s.t.} \quad p_x x + p_y y = M$$

Lagrangian function:

$$\mathcal{L}(x, y, \lambda) = U(x, y) - \lambda(p_x x + p_y y - M)$$

where λ is the **Lagrange multiplier**.

First-order conditions (FOC):

$$\frac{\partial \mathcal{L}}{\partial x} = MU_x - \lambda p_x = 0 \quad \Rightarrow \quad MU_x = \lambda p_x$$

$$\frac{\partial \mathcal{L}}{\partial y} = MU_y - \lambda p_y = 0 \quad \Rightarrow \quad MU_y = \lambda p_y$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = M - p_x x - p_y y = 0 \quad \Rightarrow \quad p_x x + p_y y = M$$

System of 3 equations in 3 unknowns: (x, y, λ)

Deriving the Tangency Condition from the Lagrangian

From the FOC:

$$MU_x = \lambda p_x \quad \text{and} \quad MU_y = \lambda p_y$$

Dividing the first by the second:

$$\frac{MU_x}{MU_y} = \frac{\lambda p_x}{\lambda p_y} = \frac{p_x}{p_y}$$

→ We obtain the tangency condition:

$$MRS = \frac{MU_x}{MU_y} = \frac{p_x}{p_y} \quad \checkmark$$

Interpretation of λ :

From $MU_x = \lambda p_x \rightarrow \lambda = MU_x / p_x$

- λ = marginal utility of income
- How much U rises if I have €1 more to spend
- At the optimum: $\lambda > 0$ (the constraint is “binding”: I would like more income)

Example: Lagrangian with Cobb-Douglas (1)

$$U(x, y) = x^{0.5}y^{0.5}, \text{ constraint } 10x + 5y = 100$$

Lagrangian:

$$\mathcal{L} = x^{0.5}y^{0.5} - \lambda(10x + 5y - 100)$$

FOC:

$$\frac{\partial \mathcal{L}}{\partial x} = 0.5x^{-0.5}y^{0.5} - 10\lambda = 0$$

$$\frac{\partial \mathcal{L}}{\partial y} = 0.5x^{0.5}y^{-0.5} - 5\lambda = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 100 - 10x - 5y = 0$$

Example: Lagrangian with Cobb-Douglas (2)

From the first two:

$$0.5 \left(\frac{y}{x} \right)^{0.5} = 10\lambda, \quad 0.5 \left(\frac{x}{y} \right)^{0.5} = 5\lambda$$

Dividing:

$$\frac{\left(\frac{y}{x} \right)^{0.5}}{\left(\frac{x}{y} \right)^{0.5}} = \frac{10}{5} \Rightarrow \left(\frac{y}{x} \right) = 2 \Rightarrow y = 2x$$

Substituting into the constraint: $10x + 5(2x) = 100 \rightarrow 20x = 100 \rightarrow x^* = 5, y^* = 10 \checkmark$

General Formula: Cobb-Douglas

Utility function: $U(x, y) = x^\alpha y^\beta$

Constraint: $p_x x + p_y y = M$

Optimal Solution

$$x^* = \frac{\alpha}{\alpha + \beta} \cdot \frac{M}{p_x}$$

$$y^* = \frac{\beta}{\alpha + \beta} \cdot \frac{M}{p_y}$$

Interpretation:

- Fraction $\alpha/(\alpha + \beta)$ of income spent on x
- Fraction $\beta/(\alpha + \beta)$ of income spent on y
- Expenditure shares depend only on α, β (not on prices!)

Special case: If $\alpha = \beta$ (e.g., $U = xy$):

$$x^* = \frac{M}{2p_x}, \quad y^* = \frac{M}{2p_y} \quad \longrightarrow \quad \text{half of income on each good}$$

Case $\alpha = \beta = 0.5$

$$U = x^{0.5}y^{0.5}, p_x = 10, p_y = 5, M = 100$$

General formula:

$$x^* = \frac{0.5}{0.5 + 0.5} \cdot \frac{100}{10} = \frac{1}{2} \cdot 10 = 5$$

$$y^* = \frac{0.5}{0.5 + 0.5} \cdot \frac{100}{5} = \frac{1}{2} \cdot 20 = 10$$

Checking the constraint:

$$10 \cdot 5 + 5 \cdot 10 = 50 + 50 = 100 \quad \checkmark$$

Checking tangency:

$$MU_x = 0.5 x^{-0.5} y^{0.5}, \quad MU_y = 0.5 x^{0.5} y^{-0.5}$$

$$MRS = \frac{MU_x}{MU_y} = \frac{0.5}{0.5} \cdot \frac{y}{x} = \frac{y}{x} = \frac{10}{5} = 2$$

$$\frac{p_x}{p_y} = \frac{10}{5} = 2 \quad \checkmark$$

$MRS = p_x/p_y \rightarrow$ tangency condition satisfied!

Exercise 1

Problem: $U(x, y) = x^{0.6}y^{0.4}$, $p_x = 4$, $p_y = 2$, $M = 120$

a: Find the optimal choice (x^*, y^*) .

Solution:

$$x^* = \frac{0.6}{0.6 + 0.4} \cdot \frac{120}{4} = 0.6 \cdot 30 = 18$$
$$y^* = \frac{0.4}{1} \cdot \frac{120}{2} = 0.4 \cdot 60 = 24$$

b: Check that the constraint is satisfied.

Solution: $4 \cdot 18 + 2 \cdot 24 = 72 + 48 = 120 \checkmark$

c: Compute the MRS at the optimum and check tangency.

Solution:

$$MRS = \frac{MU_x}{MU_y} = \frac{0.6}{0.4} \cdot \frac{y}{x} = 1.5 \cdot \frac{24}{18} = 1.5 \cdot \frac{4}{3} = 2$$
$$\frac{p_x}{p_y} = \frac{4}{2} = 2 \quad \checkmark$$

Exercise 2

Problem: $U(x, y) = \ln x + 2 \ln y$, $p_x = 3$, $p_y = 6$, $M = 90$

Note: $\ln x + 2 \ln y = \ln x + \ln y^2 = \ln(xy^2) \rightarrow$ Equivalent to $U = x^1 y^2$ (monotonic transformation)

Question: Find (x^*, y^*) .

Solution: I identify $\alpha = 1$, $\beta = 2$:

$$x^* = \frac{1}{1+2} \cdot \frac{90}{3} = \frac{1}{3} \cdot 30 = 10$$

$$y^* = \frac{2}{3} \cdot \frac{90}{6} = \frac{2}{3} \cdot 15 = 10$$

Check: $3 \cdot 10 + 6 \cdot 10 = 30 + 60 = 90 \checkmark$

Interpreting the expenditure shares:

- Expenditure on x : $3 \cdot 10 = 30$ (1/3 of income)
- Expenditure on y : $6 \cdot 10 = 60$ (2/3 of income)
- Shares reflect the weights $\alpha = 1$, $\beta = 2$

Special Cases: Perfect Substitutes and Complements

Perfect substitutes: $U(x, y) = ax + by$

- Constant MRS: $MRS = a/b$ (always)
- Comparison with p_x/p_y :
 - ▶ If $a/b > p_x/p_y$: Optimum: buy only x (corner solution)
 - ▶ If $a/b < p_x/p_y$: Optimum: buy only y (corner solution)
 - ▶ If $a/b = p_x/p_y$: Indifference (any point on the line)

Perfect complements: $U(x, y) = \min\{ax, by\}$

- Optimum always at the kink: $ax^* = by^*$ (fixed proportion)
- Solving with the constraint:

$$x^* = \frac{bM}{bp_x + ap_y}, \quad y^* = \frac{aM}{bp_x + ap_y}$$

Note: In these cases, the standard Lagrangian method does not work (non-differentiable function or corner solution). We use economic reasoning + the constraint.

Corner Solutions: When the Optimum Is Extreme

So far: Interior solutions with tangency ($MRS = p_x/p_y$)

But sometimes: The optimal solution lies at an **endpoint** of the constraint!

Corner Solution

The consumer spends all her income on a **single good** (zero quantity of the other)

When does it happen?

1. **Perfect substitutes** with $MRS \neq p_x/p_y$
2. **Very unbalanced preferences** (one good much more desired)
3. **Extreme relative prices** (one good too expensive relative to its utility)

Feature: The indifference curve is **not** tangent to the line, but “touches” it at an endpoint (intercept)

Example: Perfect Substitutes

Problem: $U(x, y) = 2x + 3y$ (perfect substitutes)

Constraint: $4x + 6y = 120$

Data: $p_x = 4 \text{ €}$, $p_y = 6 \text{ €}$, $M = 120 \text{ €}$

Step 1: Compute the MRS

$$MRS = \frac{MU_x}{MU_y} = \frac{2}{3} \approx 0.67$$

Step 2: Compute the price ratio

$$\frac{p_x}{p_y} = \frac{4}{6} = \frac{2}{3} \approx 0.67$$

Step 3: Compare

$$MRS = \frac{2}{3} = \frac{p_x}{p_y} \Rightarrow \text{Indifference!}$$

Solution

The consumer is **indifferent** among all bundles on the budget line.

Optimal solutions: **all points** from $(30, 0)$ to $(0, 20)$ are equally good!

Example: Pure Corner Solution (1)

Problem: $U(x, y) = 3x + 2y$ (perfect substitutes)

Constraint: $5x + 4y = 100$

Data: $p_x = 5 \text{ €}$, $p_y = 4 \text{ €}$, $M = 100 \text{ €}$

Step 1: Compute the MRS

$$MRS = \frac{MU_x}{MU_y} = \frac{3}{2} = 1.5$$

Step 2: Compute the price ratio

$$\frac{p_x}{p_y} = \frac{5}{4} = 1.25$$

Step 3: Compare

$$MRS = 1.5 > \frac{p_x}{p_y} = 1.25$$

Example: Pure Corner Solution (2)

Interpretation:

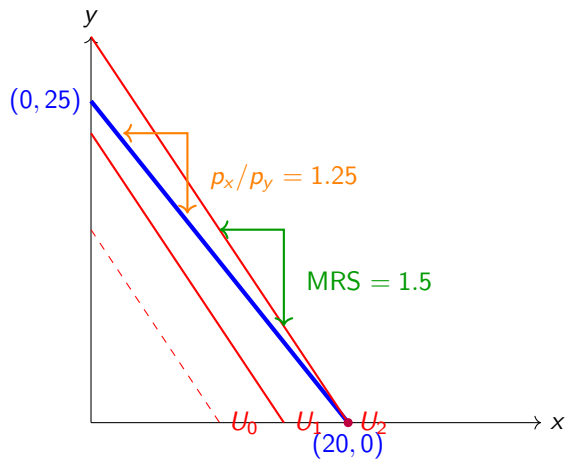
- The consumer values x more than its relative market price
- For her: 1 unit of x is worth 1.5 units of y
- But in the market: 1 unit of x costs only 1.25 units of y
- **Best option:** Buy only x !

Optimal Solution: Corner

$$x^* = \frac{M}{p_x} = \frac{100}{5} = 20, \quad y^* = 0$$

Graph: Corner Solution

Case: $MRS = 1.5 > p_x/p_y = 1.25$



Curves steeper than the constraint $\rightarrow$ corner solution in x

Opposite Case: Buy Only y

Problem: $U(x, y) = x + 3y$ (perfect substitutes)

Constraint: $4x + 2y = 80$

Calculations:

$$MRS = \frac{MU_x}{MU_y} = \frac{1}{3} \approx 0.33$$

$$\frac{p_x}{p_y} = \frac{4}{2} = 2$$

Comparison:

$$MRS = 0.33 < \frac{p_x}{p_y} = 2$$

Interpretation:

- The consumer values x **less** than its market price
- For her: 1 unit of x is worth only 0.33 units of y
- But in the market: 1 unit of x costs 2 units of y
- **Best option:** Buy only y !

Rule of Thumb: Corner Solutions (1)

For perfect substitutes $U = ax + by$:

Decision Rule

1. Compute $MRS = a/b$
2. Compute p_x/p_y
3. Compare:
 - ▶ If $\frac{a}{b} > \frac{p_x}{p_y} \rightarrow$ Buy only x : $(M/p_x, 0)$
 - ▶ If $\frac{a}{b} < \frac{p_x}{p_y} \rightarrow$ Buy only y : $(0, M/p_y)$
 - ▶ If $\frac{a}{b} = \frac{p_x}{p_y} \rightarrow$ Indifference (any point on the constraint)

Rule of Thumb: Corner Solutions (2)

Economic intuition:

- Choose the good with the **best utility/price ratio**
- $MRS > p_x/p_y$: x gives more “bang for the buck”
- $MRS < p_x/p_y$: y gives more “bang for the buck”

Note: With strictly convex preferences (Cobb-Douglas, etc.), corner solutions are rare—they require additional constraints (e.g., $x, y \geq 0$ binding for other reasons).

Exercise: Identify the Type of Solution

For each case, determine whether the solution is interior or a corner solution, and find the optimum.

a) $U = 4x + 2y$, constraint $3x + 6y = 90$

Solution: $MRS = 4/2 = 2$, $p_x/p_y = 3/6 = 0.5$

$MRS = 2 > 0.5 \rightarrow$ Corner: $(x^*, y^*) = (30, 0) \checkmark$

b) $U = x + 4y$, constraint $2x + 3y = 60$

Solution: $MRS = 1/4 = 0.25$, $p_x/p_y = 2/3 \approx 0.67$

$MRS = 0.25 < 0.67 \rightarrow$ Corner: $(x^*, y^*) = (0, 20) \checkmark$

c) $U = x^{0.5}y^{0.5}$, constraint $4x + 2y = 80$

Solution: Cobb-Douglas $\rightarrow$ **Interior** solution (always with Cobb-Douglas)

Formula: $x^* = 0.5 \cdot 80/4 = 10$, $y^* = 0.5 \cdot 80/2 = 20 \checkmark$

d) $U = 3x + 3y$, constraint $2x + 2y = 100$

Solution: $MRS = 3/3 = 1$, $p_x/p_y = 2/2 = 1$

$MRS = p_x/p_y \rightarrow$ Complete **indifference**: any point from $(50, 0)$ to $(0, 50) \checkmark$

Overall Summary

Part 1: Budget Constraint

- Equation: $p_x x + p_y y = M$
- Line with slope $-p_x/p_y$ (market trade-off)
- Budget set: affordable bundles

Part 2: Changes

- M changes $\rightarrow$ parallel shift
- Prices change $\rightarrow$ rotation
- Proportional inflation $\rightarrow$ no real effect

Part 3: Optimal Choice

- Condition: $MRS = p_x/p_y$ (tangency)
- Methods: substitution and Lagrangian
- Cobb-Douglas formula: $x^* = \frac{\alpha}{\alpha+\beta} \frac{M}{p_x}$

Skills Acquired

- ✓ Write and draw the budget constraint
- ✓ Analyze the effects of changes in prices/income
- ✓ Set up the consumer's maximization problem
- ✓ Apply the substitution and Lagrangian methods
- ✓ Compute the optimal choice for Cobb-Douglas
- ✓ Check the tangency condition

Optional Homework Exercises

1. $U = x^{0.4}y^{0.6}$, $p_x = 2$, $p_y = 4$, $M = 80$. Find (x^*, y^*) .
2. Same problem. Now M rises to 120. Find the new optimum.
3. $U = \ln x + \ln y$, $p_x = 3$, $p_y = 6$, $M = 90$. Find (x^*, y^*) using the Lagrangian.
4. Verify that with $U = xy$, if M doubles, (x^*, y^*) also double.
5. Draw the budget line for $2x + 4y = 100$. Then graph the effect of:
 - a) M rises to 150
 - b) p_x doubles
 - c) 50% inflation on everything